Motion of a charge (q) in combined electric ($\vec{E}$) and magnetic field ($\vec{B}$)

(1) If a charge 'q' moves with a velocity $\vec{v}$ in presence of both electric ($\vec{E}$) and magnetic fields ($\vec{B}$), it experiences a force.

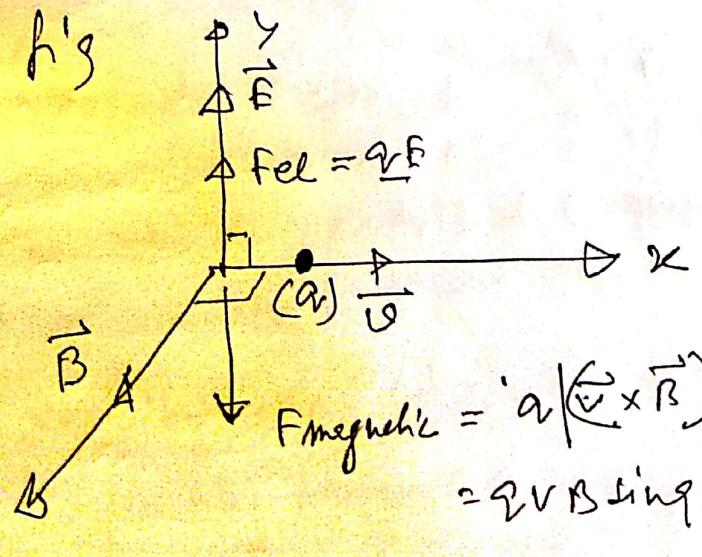
$$\vec{F} = \vec{F}_{electric} + \vec{F}_{magnetic}$$

$$\boxed{\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})}$$

This force is called Lorentz force.

(2) We consider a simple case in which electric field $\vec{E}$ & magnetic field $\vec{B}$ are perpendicular to each other and also perpendicular to velocity $\vec{v}$ of the charge q.

as in fig



$F_{magnetic} = q|(\vec{v} \times \vec{B})|$

$= qvB \sin 90^\circ$

$= qvB.$

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We see that in this situation, electric force ($F_{el} = qE$) & magnetic force $F_{mag} = qVB$ are in opposite direction.

Now by adjusting 'E' & 'B', $F_{el} = qE$ & $F_{mag} = qVB$ are made equal.

Then net force on charge is zero, & the charge move undeflected in the fields E & B.

This happen when

$$F_{el} = F_{mag}$$

i.e. $qE = qVB$

$$\boxed{V = \frac{E}{B}}$$

So when a beam of charged particles with different speed move thro' crossed electric & magnetic field (i.e. $\vec{E} \perp \vec{B}$), only those charged particles with speed $\frac{E}{B}$ pass undeflected through the region of crossed electric & magnetic field.

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ex! Problem (1) A proton moves at a constant velocity of 50 m/s along x axis, in uniform electric and magnetic field. Magnetic field is $\vec{B} = (2\,mT)\hat{k}$. What is electric field.

$$\vec{F}_{net} = q\vec{E} + q(\vec{v} \times \vec{B})$$

$\because \vec{F}_{net} = \vec{0}$, so $\vec{E} = -(\vec{v} \times \vec{B})$

or, $\vec{E} = -[50\hat{i} \times 2 \times 10^{-3}\hat{k}]$ (Board + NEET + WBJEE)

or, $\vec{E} = -0.1\hat{k}\ V m^{-1}$

~~ex 1, A charged particle of charge q~~

problem ex 2 A charged particle moves undeflected through a region of space containing mutually perpendicular and constant electric and magnetic fields 'E' and 'B'. When the electric field is switched off, the particle moves in a circular path of radius 'R'. Find specific charge (charge/mass) of the particle? (JEE/NEET; Zenel)

Case (i) As the charged particle (q, m) moves undeflected in crossed electric (E) & magnetic field (B), so $[\vec{v} \perp \vec{E} \perp \vec{B}]$

So, magnetic force = electric force

$$F_{mag} = F_{el}$$

$$qvB = qE$$

$$\boxed{v = \frac{E}{B}}$$

Case (ii) $E = 0$ & $\vec{v} \perp \vec{B}$, so magnetic force on charge is qvB & it provides centripetal force to charge.

$$qvB = \frac{mv^2}{R}$$

$$\therefore \frac{q}{m} = \frac{v}{BR}$$

$$\frac{q}{m} \text{ (sp charge)} = \frac{1}{BR} \times \frac{E}{B}$$

$$\therefore \frac{q}{m} = \frac{E}{B^2 R}$$

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